

Aligned but Not Retrieved: Target-Language Selection Failure in Bilingual Contrastive Fine-Tuning

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Motivation

Multilingual retrieval-augmented generation (RAG) systems face a failure mode we call language drift: a user queries in English targeting a Spanish knowledge base, but the retriever—a sentence encoder fine-tuned with contrastive learning—returns French or German documents instead, because semantically equivalent translations are indistinguishable in embedding space, and the language model generates in the wrong language. This failure is documented in production (Shi et al., 2025) and currently patched with decoder-side workarounds. We trace the root cause to the retriever and show that a training-time fix at the encoder eliminates the need for those patches.

Multilingual models such as XLM-RoBERTa (Conneau et al., 2020) learn partially aligned cross-lingual representations, and contrastive fine-tuning with InfoNCE (van den Oord et al., 2018) further strengthens this alignment by pulling translation pairs together while separating non-parallel sentences. Standard retrieval benchmarks such as Tatoeba (Artetxe and Schwenk, 2019a) and BUCC (Zweigenbaum et al., 2017) evaluate performance by presenting English queries against a *monolingual* candidate pool—for example, only Spanish sentences. In this closed-pool setting, semantic alignment and target-language selection are perfectly correlated by construction: any retrieved sentence is simultaneously the right meaning and the right language. The two objectives are indistinguishable.

Our experiments expose a gap concealed by this design. After 2000 steps of InfoNCE fine-tuning on English–Spanish (EN–ES) bitext from OPUS-100, the model achieves $P@1 = 0.994$ on the standard closed pool of 1012 Spanish candidates. When we expand the candidate pool to 5060 sentences spanning five languages (ES, FR, DE, SW, AR), all drawn from FLORES-200 devtest and therefore semantically aligned by construction (each language contains a translation of the same 1012 source sentences), $P@1$ drops to 0.382. We decompose these failures into *wrong-language true positives* (the model retrieves the semantically correct sentence but in the wrong language) and *true errors* (the model retrieves an unrelated sentence). The result is unambiguous: $P@1$ -any, where any correct translation in any language counts as a hit, equals 1.000 for both the encoder and projected spaces. True errors are exactly zero. Every failure is a wrong-language true positive.

Language confusion is not uniform and follows a clear typological proximity gradient. Portuguese, the closest language to Spanish, produces the worst pairwise result (ES+PT: $P@1 = 0.364$)—worse than the full five-language pool—because all confusion concentrates on a single competitor. French and German together account for 96% of hard-pool failures (ES+FR: 0.567, ES+DE: 0.542), while Swahili and Arabic are near the easy-pool ceiling (ES+SW: 0.936, ES+AR: 0.902). Notably, German is Germanic, not Romance, yet confuses nearly as badly as French; the relevant factor is surface-level similarity (shared script, vocabulary overlap, syntactic proximity), not family membership.

This failure is specific to multiway-parallel evaluation and invisible on standard benchmarks. When we repeat the hard-pool test on OPUS-100 test data, where each language pair draws from independent source documents and wrong-language hits are genuine retrieval errors, the gap drops from 0.612 to 0.013. The model performs normally when candidate languages do not share source documents. The failure emerges only when semantically identical sentences in competing languages are present as candidates—precisely the multiway-parallel setting.

Post-hoc geometric interventions make the hard pool *worse*, not better: whitening costs 0.039 P@1 points, margin-based scoring (Artetxe and Schwenk, 2019b) costs 0.020, and both together cost 0.060. Hubness statistics confirm there is no geometry pathology to fix (hub fraction = 0.000; N_k standard deviation = 1.26 against an expected value of 2.0). The InfoNCE projection head compounds the problem: the gap in the 256-dim projected space (0.667) exceeds the gap in the 768-dim encoder space (0.612), meaning the training objective actively degrades language discrimination rather than merely ignoring it. Finally, more InfoNCE training worsens target-language selectivity monotonically: P@1-target drops from 0.382 at 2K steps to 0.259 at 10K before partially recovering to 0.297 at 20K, ruling out undertraining as an explanation and showing that the semantic collapse InfoNCE induces is progressive.

Proposed Research

We propose a diagnostic and interventional study of target-language selection failure in bilingual contrastive fine-tuning. Our central thesis is that bilingual InfoNCE optimizes semantic alignment but leaves target-language specificity underdetermined in multiway-parallel candidate pools; that this failure is invisible to standard closed-pool evaluation, follows a typological proximity gradient, and is addressable with a training-time language-conditioning signal.

First, we *characterize the failure across training pairs*. Beyond the EN–ES pilot, we fine-tune on EN–FR, EN–DE, and EN–SW and evaluate each against the five-language hard pool. We test whether confusion severity scales with typological proximity between the trained language and its neighbors, and whether training on a typologically isolated pair (EN–SW) produces less cross-lingual confusion than a crowded one (EN–ES).

Second, we *evaluate two training-time interventions*. (i) *Language-aware hard negative mining*: we inject same-source wrong-language translations directly into the InfoNCE denominator using FLORES-200 dev pairs, so the model explicitly learns during EN–ES training that semantically equivalent French and German sentences are wrong answers. This requires no architectural change. Hard negative mining closes 99.7% of the FLORES gap in 2000 steps. However, the bystander evaluation reveals a severe limitation: by globally pushing all non-target-language representations away from EN queries during training, the model collapses EN→FR, EN→DE, and EN→SW easy-pool P@1 to under 2%—a catastrophic regression for any system that serves more than one target language. Standard bilingual EN→ES retrieval also regresses by approximately 10% on OPUS-100 test. Hard negative mining is therefore viable only for fixed-target-language systems, not as a general multilingual retrieval fix. (ii) *Language-conditioned query encoder*: motivated directly by the bystander collapse above, we add a learned per-target-language embedding to the query’s mean-pooled representation before the projection head, combined with an auxiliary contrastive loss over six-tuples (EN, ES, FR, DE, SW, AR) from FLORES-200 dev that pulls queries toward their target-language translations and pushes away wrong-language translations of the same sentence. At inference, the query is conditioned on the desired target language—a targeted pull rather than a global push, which is the mechanism required to avoid bystander degradation. We sweep auxiliary loss weight $\lambda \in \{0.1, 0.5, 1.0, 2.0\}$. Because the failure is at least as severe in the raw encoder space as in the projected space, language conditioning is applied at the encoder output level to affect the full projection path.

We compare both interventions against: pretrained XLM-R (zero-shot), LaBSE (Feng et al., 2022) as a zero-shot baseline trained to avoid this collapse, vanilla InfoNCE at the 2K-step best checkpoint, oracle language filter (ceiling), and predicted language filter (langdetect). We report P@1-target, P@1-any, and the per-language confusion matrix for all conditions, on both the training pair (EN–

ES) and bystander languages.

Third, we *validate the retrieval fix end-to-end in a RAG pipeline*. Using MKQA or XQuAD with a multilingual Wikipedia passage corpus (ES, FR, DE, SW, AR), we demonstrate that the training-time interventions above directly reduce language drift in generated outputs—confirming that fixing the encoder eliminates the need for decoder-side workarounds. We measure retrieval language accuracy (fraction of top-1 passages in the target language) and answer language drift (fraction of LLM outputs in the wrong language).

Related Work

XLm-RoBERTa (Conneau et al., 2020) is our base model. InfoNCE (van den Oord et al., 2018) and SimCSE (Gao et al., 2021) form the fine-tuning backbone. LaBSE (Feng et al., 2022) achieves strong cross-lingual retrieval results but trains and evaluates on closed monolingual pools—the design that conceals the failure mode we study. Artetxe and Schwenk (2019b) introduce the margin scoring function we test as a post-hoc baseline, and their LASER system (2019a) establishes the Tatoeba benchmark on which we build.

Shi et al. (2025) document language drift in production multilingual RAG systems—generated output appearing in the wrong language—and propose decoder-side workarounds; we instead identify and fix the root cause at the retriever.

Huang et al. (2025) show that joint cross-lingual alignment and language-specialization training cause interference and that separate modular training resolves it—directly validating our failure mode from a complementary angle and motivating our language-conditioned architecture. Asai et al. (2022) demonstrate that same-meaning different-language translation pairs function as effective hard negatives for multilingual dense retrieval, supporting our Phase 2a approach.

The closest related work is LAReQA (Roy et al., 2020), which introduces language-agnostic answer retrieval from a multilingual pool and explicitly notes that closed-pool benchmarks do not test language confusion bias—directly corroborating our diagnosis. However, LAReQA optimizes the *opposite* objective: it rewards retrieving the best answer *regardless* of language. Our setting requires language-targeted retrieval from a multilingual pool, where semantically correct translations in the wrong language are failures, not successes. To our knowledge no prior work identifies the gap between P@1-any and P@1-target as a distinct failure mode of bilingual contrastive training, characterizes its typological proximity gradient, demonstrates that it vanishes on non-parallel benchmarks, or proposes a training objective that addresses it without collapsing retrieval in bystander languages. The bystander collapse produced by hard negative mining—a known technique—makes the language-conditioned query encoder the central architectural contribution of this work.

Evaluation

We evaluate retrieval using P@1-target, P@1-any, wrong-language positive rate, true error rate, and the full per-language confusion matrix under both closed (1012 same-language candidates) and hard (5060 five-language candidates) pool settings, using FLORES-200 devtest throughout. We additionally report pairwise two-language pool results (ES+PT, ES+FR, ES+DE, ES+SW, ES+AR) to quantify the proximity gradient and verify that interventions reduce confusion for the most confusable pairs without degrading the already-strong SW and AR results. Critically, we evaluate all interventions on *bystander easy pools*—EN→FR, EN→DE, EN→SW, EN→AR retrieval on single-language candidate pools—to characterize whether each fix degrades general multilingual retrieval capability. Hard negative mining collapses bystander P@1 to under 2%; the language-conditioned encoder is specifically designed to avoid this by conditioning queries on the target

language at inference time rather than globally repelling bystander languages during training.

We compare FLORES-200 devtest results against OPUS-100 test results to demonstrate the benchmark design effect: the gap invisible on OPUS-100 (0.013) grows to 0.612 on FLORES, establishing that multiway-parallel evaluation is necessary to observe the failure and that the finding is not a generic retrieval failure.

For geometric analysis we track layer-wise CKA and Procrustes similarity before and after fine-tuning to verify that language conditioning does not degrade the underlying cross-lingual alignment that standard benchmarks measure. All retrieval experiments are run on the training pair (EN–ES) and on bystander languages (FR, DE, SW, AR) to confirm that interventions do not transfer confusion to previously unaffected language pairs.

Feasibility and Implementation

We build on a working codebase using XLM-R base, mean-pooled L2-normalized sentence representations, and InfoNCE loss with AdamW on OPUS-100 bitext. The hard-pool evaluation pipeline, error decomposition, per-family pool breakdown, OPUS-100 comparison, and language confusion matrix are fully implemented and produced the results reported above. Checkpoints at 2K, 5K, 10K, and 20K steps are trained and available. Multiway parallel data for the auxiliary loss is available from FLORES-200 dev (997 sentences, separate from the devtest split used for all reported metrics). The language-conditioned architecture adds one learned embedding per target language to the existing encoder output; the hard negative variant modifies only the data collator. Both interventions are feasible on a single GPU in mixed precision and ablate cleanly against the vanilla InfoNCE baseline. MKQA and XQuAD are publicly available; the RAG end-to-end evaluation requires one inference pass per encoder configuration using a publicly available LLM (GPT-4o mini).

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